Code-Layout, Readability and Reusability

Date	01-July-2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper Python indentation followed in all files
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into templates, static, models, notebooks, and application files
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as nitrogen, phosphorus, potassium, temperature, humidity are descriptive
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named according to their purpose such as predict_crop() and load_model()
5	Code Comments	Inline and block comments explain logic	Yes	Important sections contain comments for better understanding
6	Modular Design	Code is split into reusable functions/modules	Yes	Machine learning, preprocessing, and Flask components are separated logically
7	No Redundant Code	Duplicate or unused code is removed	Yes	

				Unused functions and duplicate code blocks were removed
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and error handling implemented

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Module Python, Pandas	Dataset cleaning and preparation	High
2	Machine Learning Model Module	Scikit-learn	Crop prediction and recommendation	High
3	Flask Backend Module	Flask, Python	Request handling and prediction processing	processing High
4	HTML Form Interface	HTML, Bootstrap	User input collection pages	Medium
5	Prediction Result Module	Flask, HTML	Displaying crop recommendations	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules
Readability	5	Meaningful variable names and clear coding style
Reusability	5	Modular design allows components to be reused easily
Documentation / Comments	4	Important sections are documented with comments
Overall Score	5	The project follows good coding practices and maintainable design